DeepSeek-V2

A Strong, Economical, and Efficient Mixture-of-Experts Language Model

TOTAL PARAMETERS	ACTIVATED PER TOKEN	CONTEXT LENGTH
236B	**21B**	**128K**

Multi-Head Latent Attention (MLA) · DeepSeekMoE · GRPO Alignment

arXiv: 2405.04434 · DeepSeek-AI · Pretrained on 8.1T tokens

MOTIVATION

The LLM Scaling Dilemma: Performance vs. Cost and Inference Speed

Larger models yield better intelligence but incur prohibitive training costs and KV cache bottlenecks — demanding a new architecture paradigm.

Scaling Delivers Intelligence

More parameters → stronger language tasks

- MMLU, reasoning, and code benchmarks all improve with scale
- Industry trajectory: 7B → 70B → 100B+
- Open-source frontier driven by scale

But Training Cost Explodes

Dense scaling grows cost super-linearly

- Training FLOPs ∝ N^2 for dense models
- KV cache grows O(n) per token: memory bottleneck at inference
- Throughput degrades with context length

Solution: Sparse Activation + Efficient Attn.

MoE + MLA to decouple capacity from cost

- Mixture-of-Experts: activate only a fraction of parameters per token
- MLA: compress KV cache via low-rank projection
- DeepSeek-V2 embodies both innovations

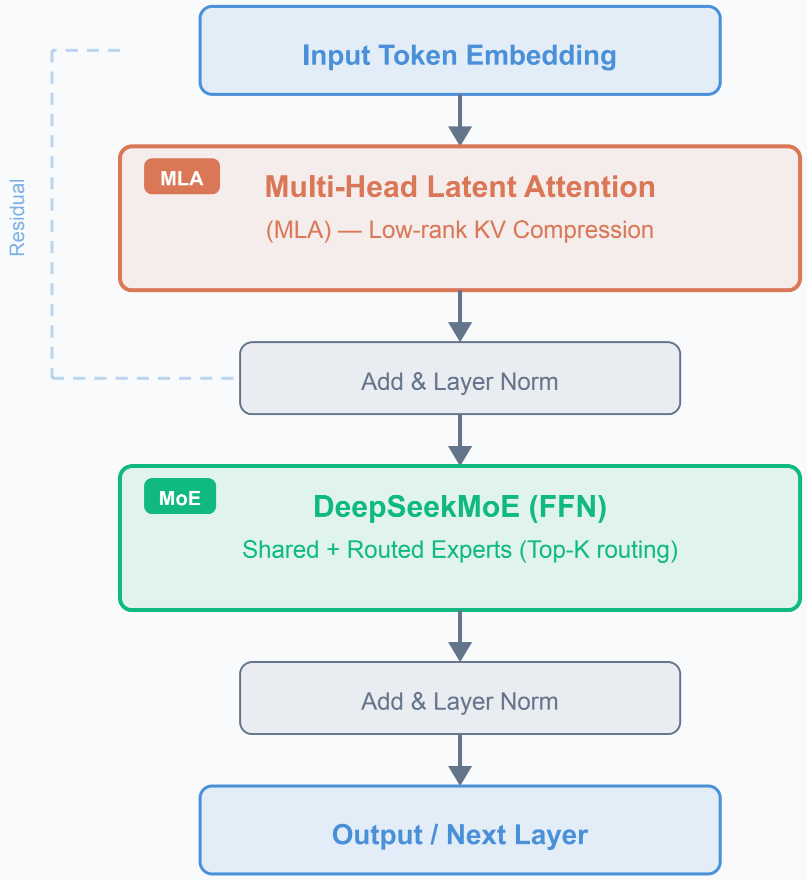
Source: DeepSeek-AI, arXiv:2405.04434

ARCHITECTURE

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in Every Transformer Block

236B total / 21B activated per token — efficient attention (MLA) replaces MHA; sparse FFN (DeepSeekMoE) replaces dense FFN.

Transformer Block Architecture



Model Specifications

SCALE
236B total params · Only 21B activated per token
MoE architecture decouples capacity from activation cost

CONTEXT
128K-token context window
Enabled by MLA's compact KV cache representation

ATTENTION
MLA replaces Multi-Head Attention
Low-rank KV compression → 93.3% smaller KV cache

FFN LAYER
DeepSeekMoE replaces dense FFN
Shared + routed experts, top-K routing, device-limited

PRETRAINING
8.1T tokens — multi-source + extended Chinese
SFT on 1.5M sessions · GRPO-based RL alignment

EFFICIENCY GAINS vs. DEEPSEE K 67B
42.5% cost · 93.3% KV cache · 5.76× throughput

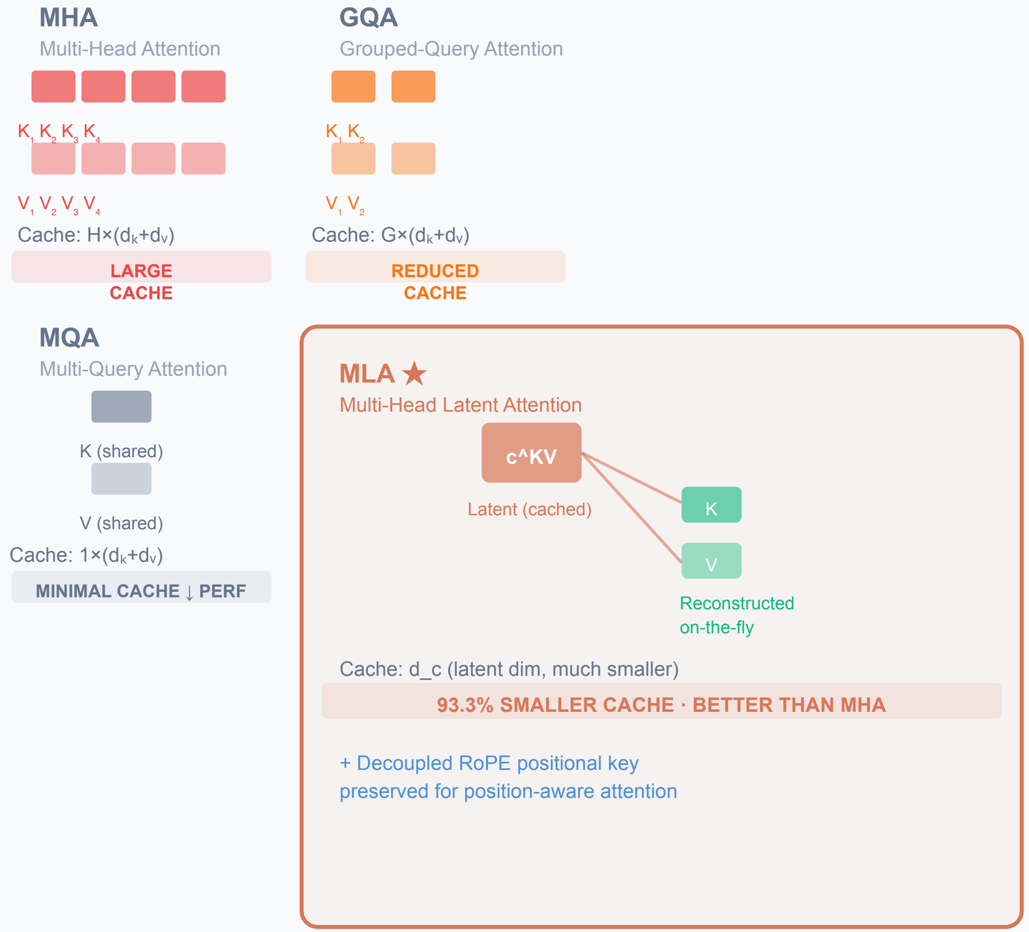
Source: DeepSeek-AI, arXiv:2405.04434

INNOVATION · MLA

Multi-Head Latent Attention Compresses KV Cache by 93.3%

MLA's low-rank KV joint compression outperforms MHA quality while requiring drastically less memory — making GQA/MQA obsolete.

Attention Mechanism Comparison



How MLA Works

1. **Problem: KV Cache Bottleneck**
   Full K,V per head per token = O(H×L) memory
   Limits batch size and context length at inference

2. **MLA: Low-Rank Joint KV Compression**
   Compress K and V jointly into compact latent vector c^KV via learned projection W^DKV

3. **Cache Only the Latent Vector**
   At inference: store only c^KV (dimension d_c)
   Much smaller than full K,V pairs per head

4. **Reconstruct On-the-Fly**
   Full K, V reconstructed via up-projection matrices
   + Decoupled RoPE positional key for positional info

5. **Result: Better Than MHA**
   KV cache ↓93.3% vs. MHA at same quality
   Outperforms GQA and MQA on benchmarks

Key Result
MLA = strictly better than MHA in quality,
with only 6.7% of the original KV cache size.

Source: DeepSeek-AI, arXiv:2405.04434

INNOVATION · MOE

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Fine-grained expert segmentation + shared expert isolation enables 236B-scale training at 42.5% lower cost than a 67B dense model.

Shared Experts
(N_s) — Always Active

Role: Universal knowledge
Activated for every token, regardless of routing decision.

Function: Common patterns
Captures cross-domain knowledge that benefits all tokens.

Design insight
Prevents shared information from fragmenting across routed experts unnecessarily.

Routed Experts
(N_r) — Sparsely Activated

Routing: Top-K_r selection
Token-wise gating selects the top-K_r most relevant experts.

Fine-grained segmentation
More, smaller experts increase specialization potential per expert.

Result: Only 21B active
Out of 236B total parameters, only 21B fire per token at inference — 9% of total.

Efficiency
Mechanisms

Device-limited routing
Each token uses experts on ≤M devices → cuts cross-node comm.

Auxiliary load-balance loss
Prevents expert collapse; enforces balanced utilization across GPUs.

Token-dropping (training only)
Prevents expert overflow during training without affecting inference quality or routing logic.

Key Insight
Fine-grained expert segmentation increases each expert's specialization potential. Shared expert isolation prevents common knowledge from unnecessarily occupying routed expert capacity — maximizing both efficiency and quality.

EFFICIENCY GAINS

42.5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

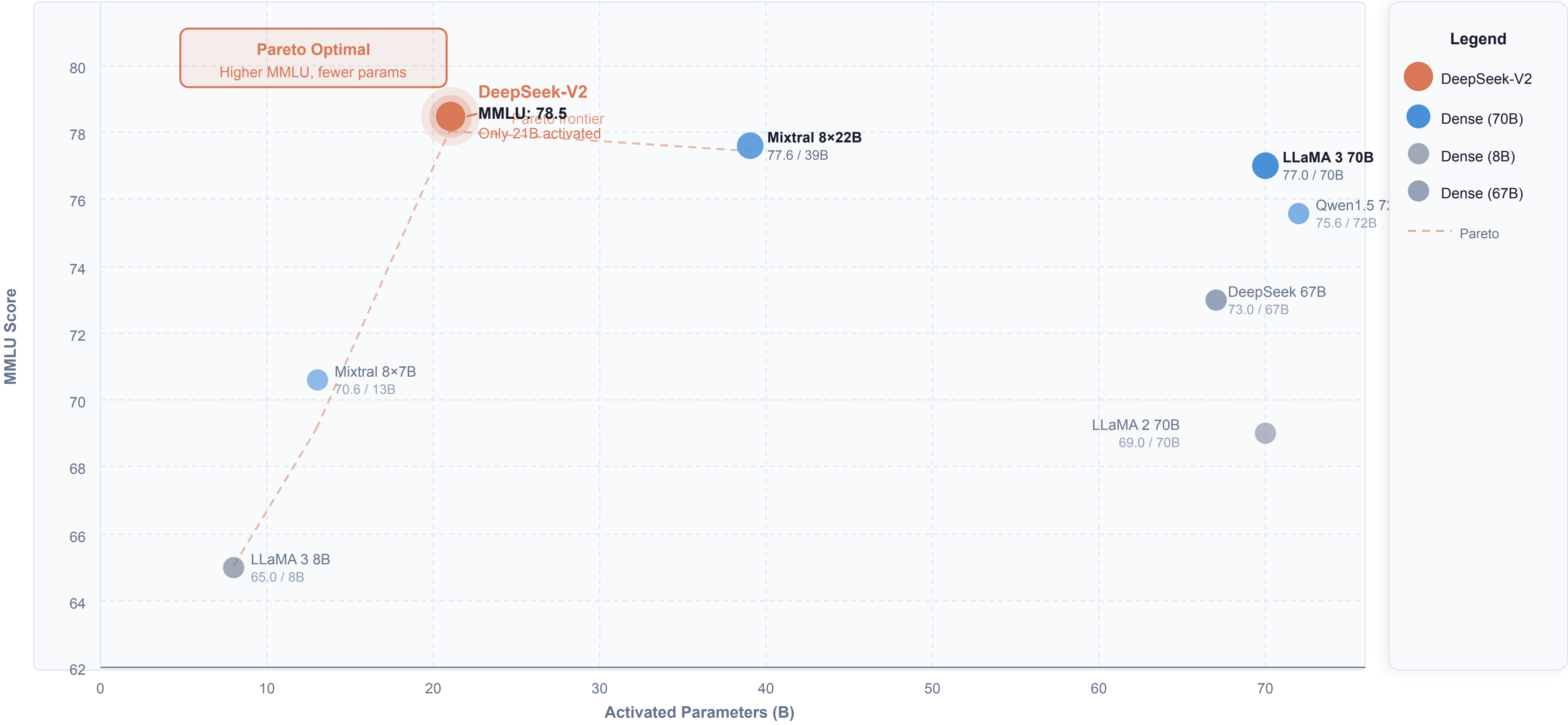
DeepSeek-V2 (MoE, 236B/21B) delivers Pareto-dominant efficiency over the DeepSeek 67B dense baseline on all three metrics.

METRIC	DeepSeek 67B Dense Baseline	DeepSeek-V2 MoE (236B / 21B activated)
Training Cost Compute cost relative to dense baseline MoE enables 236B-param model trained cheaper than 67B dense	Baseline (100%) Reference point	57.5% of baseline ↓ 42.5% Cost vs. DeepSeek 67B
KV Cache Size Memory footprint of key-value cache at inference MLA low-rank compression vs. full KV pairs in standard MHA	Baseline (100%) Full KV per head (MHA)	6.7% of baseline ↓ 93.3% Cache vs. DeepSeek 67B
Max Generation Throughput Maximum tokens/second at inference (same hardware) Enabled by smaller KV cache → larger batch sizes → higher utilization	Baseline (1×) Reference throughput	5.76× faster ↑5.76× vs. DeepSeek 67B

BENCHMAR
K

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activated Params

DeepSeek-V2 achieves 78.5 MMLU with only 21B activated parameters — outperforming 70B+ dense models while activating 3× fewer params.



Source: DeepSeek-AI, arXiv:2405.04434 · Activated parameters and MMLU scores from paper Table 3

ALIGNMENT

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

After SFT + GRPO alignment: best open-source Chinese chat model, competitive with closed-source systems in English.

Alignment Pipeline

Pretraining
8.1T tokens · multi-source · extended Chinese

↓

Supervised Fine-Tuning (SFT)
1.5M conversational sessions
Multi-turn dialogue · instruction following · safety

↓

Reinforcement Learning (GRPO)
Group Relative Policy Optimization
Preference alignment · helpfulness · harmlessness

↓

DeepSeek-V2 Chat (RL)
Strongest open-source Chinese chat model

Why GRPO?
Group Relative Policy Optimization avoids the need for a separate value model — more efficient than PPO for LLMs.
Rewards derived from group-relative comparisons of outputs.

Chat Evaluation Results — DeepSeek-V2 Chat (RL)

Benchmark	Score
AlpacaEval 2.0 Length-controlled win rate vs. GPT-4 · Open-ended English chat	38.9%
MT-Bench Multi-turn English instruction following · Scale: 1–10	8.97
AlignBench (Chinese) Comprehensive Chinese alignment benchmark · Scale: 1–10	7.91

Best Open-Source Chinese Chat Model
On AlignBench, DeepSeek-V2 Chat (RL) outperforms all open-source models and most closed-source systems.

Competitive in English Open-Ended Evaluation
AlpacaEval 38.9% win rate positions DeepSeek-V2 Chat (RL) among the top open-source models in English too.

CONCLUSION

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

MLA is a strictly better KV cache solution
93.3% smaller KV cache vs. MHA — better quality than GQA/MQA — enables 128K context

DeepSeekMoE enables economical 236B-scale training
42.5% lower training cost than DeepSeek 67B dense — shared + routed expert design maximizes capacity

21B activated params match or exceed 70B-class dense models
DeepSeek-V2 surpasses LLaMA 3 70B and Qwen1.5 72B on MMLU while activating 3× fewer parameters

5.76× throughput improvement makes large-scale deployment practical
Smaller KV cache → larger batch sizes → dramatically higher hardware utilization

DeepSeek-V2 Chat (RL): strongest open-source Chinese model
AlignBench 7.91 outperforms all open-source + most closed-source; MT-Bench 8.97; competitive in English

"Sparse activation + efficient attention is a Pareto-optimal path for LLM scaling."